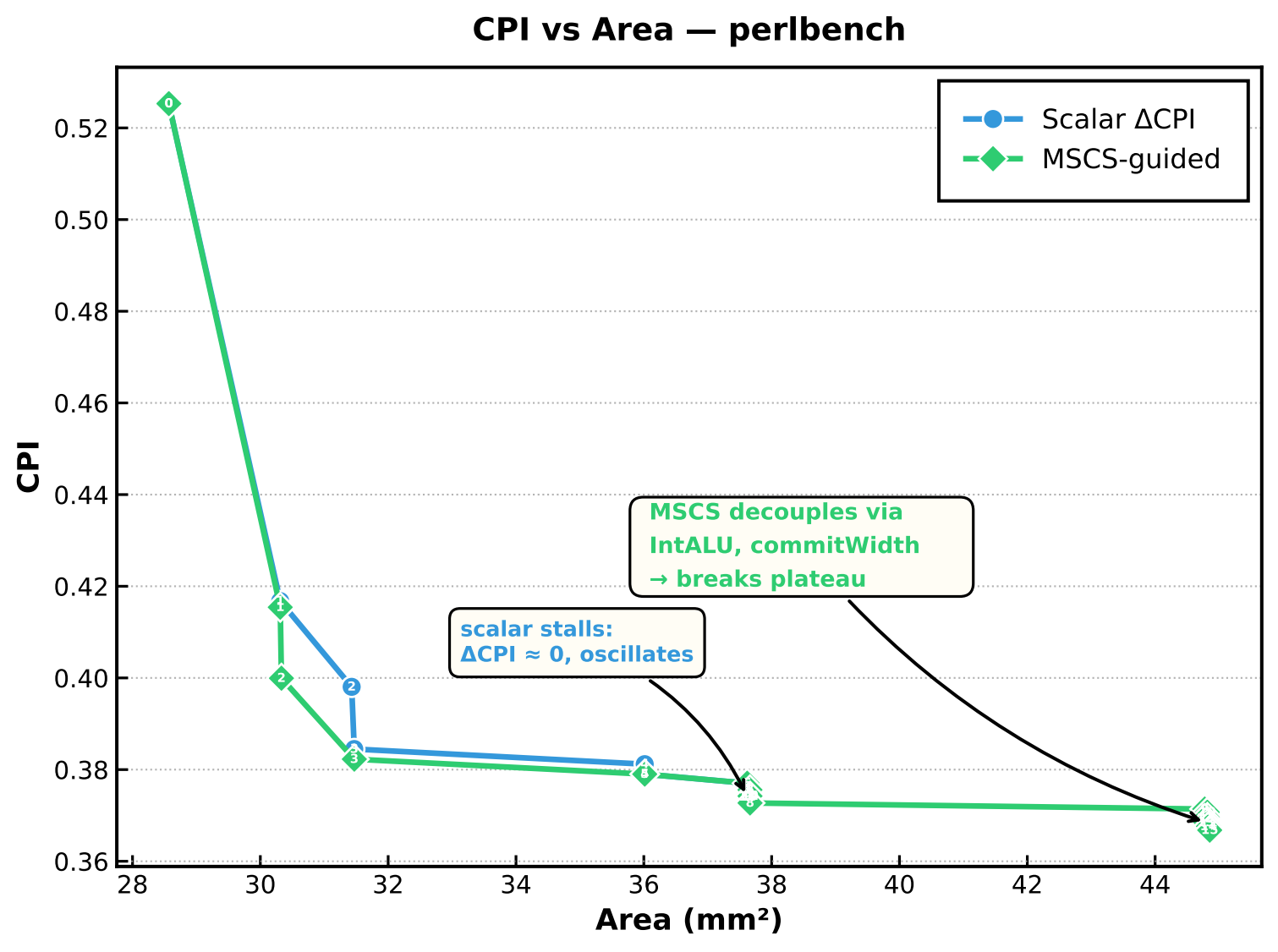
CPI vs Area — perlbench
Scalar ΔCPI
MSCS-guided
MSCS decouples via
IntALU, commitWidth
→ breaks plateau
scalar stalls:
ΔCPI ≈ 0, oscillates
CPI
0.52
0.50
0.48
0.46
0.44
0.42
0.40
0.38
0.36
28
30
32
34
36
38
40
42
44
Area (mm²)